Risk and Opportunity Management (RSK)

RSK Overview

Required PA Information

Intent

Identifies, records, analyzes, and manages potential risks or opportunities.

Value

Mitigates adverse impacts or capitalizes on positive impacts to increase the likelihood of meeting objectives.

Additional Required PA Information

The term risk refers to uncertainties that may have a negative impact on achieving objectives. The term opportunity refers to uncertainties that may have a positive impact on achieving objectives.

Explanatory PA Information

Practice Summary

Level 1

RSK 1.1	Identify and record risks or opportunities and keep them updated.

Level 2

RSK 2.1	Analyze identified risks or opportunities.
RSK 2.2	Monitor identified risks or opportunities and communicate status to affected stakeholders.

Level 3

RSK 3.1	Identify and use risk or opportunity categories.
RSK 3.2	Define and use parameters for risk or opportunity analysis and handling.
RSK 3.3	Develop and keep updated a risk or opportunity management strategy.
RSK 3.4	Develop and keep updated risk or opportunity management plans.
RSK 3.5	Manage risks or opportunities by implementing planned risk or opportunity management activities.

Additional PA Explanatory Information

Risk and opportunity management is a continuous, forward-looking process and includes:

- Identifying and mitigating potential negative impacts that may make it difficult to meet objectives

- Identifying and leveraging potential positive impacts for improving performance or progress towards achieving objectives
- All levels of an organization

Identify risks or opportunities early by working with affected stakeholders. Before dedicating resources to address risks or opportunities, determine which are worth pursuing. During risk and opportunity management, consider:

- Timeliness of the needed response
- Significance or consequences of the situation
- Regularity or chances of the situation occurring
- Internal, external, technical, and non-technical sources
- Early implementation of handling activities that may be less disruptive and costly
- Uncertainties that could most affect the ability to perform
- Industry standards and best practices that can help to identify and manage uncertainty

Risk and opportunity management involves:

- Defining a strategy
- Identifying and analyzing risks or opportunities
- Handling identified risks or opportunities includes:
    - Developing mitigation and contingency plans for risks
    - Developing plans to leverage opportunities
- Implementing risk or opportunity handling plans

Risk and opportunity management practices help organizations evolve from reactively identifying uncertainties to systematically planning, anticipating, and handling them.

Related Practice Areas

Planning (PLAN)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

As an empirical framework, agile, does not define practices and processes for risk and opportunity management. Risk and opportunity management is performed using visual information indicators, daily standups, short Sprints (iterations) with frequent feedback, and close collaboration within teams and customers. Some agile teams reduce technical risk by using “spikes,” or rapid prototypes performed early in the project. Risk and opportunity management can be easily added to the planning, execution, and retrospective activities of each Sprint, or selected Sprints.

Level 1

RSK 1.1

Required Practice Information

Practice Statement

Identify and record risks or opportunities and keep them updated.

Value

Enables organizations to avoid or minimize the impact of risks and leverage potential opportunities related to achieving objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify and describe risks or opportunities clearly.

Example Activities

Example Activities	Further Explanation
Identify the risks associated with the work.	Identify and clearly describe risks that could negatively affect work and plans. Risk identification should consider: • Cost • Schedule • Work tasks • Performance • Achievement of business objectives • Environmental concerns, e.g., weather or natural disasters, political changes, or telecommunications failures • Requirements • Technology • Staffing • Funding • Suppliers • Regulatory constraints Perform risk identification and management during planning and review activities.
Record the risks.	State the risk and the impact of its occurrence.

Example Activities	Further Explanation
Identify opportunities.	Identify and clearly describe opportunities that could positively affect work and plans. For example: • Cost • Schedule • Work tasks • Performance • Achievement of business objectives • Requirements • Technology • Staffing • Funding • Suppliers Perform opportunity identification and management during review and planning activities.
Record opportunities.	State the opportunity and its potential benefits.
Identify the affected stakeholders associated with each risk or opportunity.	

Example Work Products

Example Work Products	Further Explanation
List of identified risks or opportunities	Include context, conditions, and consequences of risks or benefits of opportunities.

Level 2

RSK 2.1

Required Practice Information

Practice Statement

Analyze identified risks or opportunities.

Value

Increases the likelihood of achieving objectives by reducing the impact of risks or leveraging opportunities.

Additional Required Information

Analysis of the risks includes potential impact, likelihood of occurrence, and the timeframe in which they are likely to occur.

Analysis of opportunities includes potential benefits, potential costs, and the timeframe for action.

Explanatory Practice Information

Additional Explanatory Information

The analysis of risks or opportunities should consider:

- Assigning mitigation or contingency to the highest priority or the most critical risks
- Leveraging opportunities by assigning the highest priority to those with the greatest benefit

Example Activities

Example Activities	Further Explanation
Analyze the identified risks.	Analyze risks to understand their effect on achieving the work's objectives. Examples of risk identification and analysis techniques include: • Assessments • Checklists • Structured interviews • Brainstorming • Failure Mode and Effects Analysis (FMEA) • Failure Mode, Effects and Criticality Analysis (FMECA) • Strength, Weakness, Opportunity, and Threat (SWOT) analysis • Poka-yoke analysis

Example Activities	Further Explanation
Review the Work Breakdown Structure (WBS), plan, and schedule for risks.	
Identify the impact of each risk.	
Identify the probability of occurrence for each risk.	
Assign priorities to each risk based on the impact and probability of occurrence.	
Analyze the identified opportunities.	Examples of opportunity identification and analysis techniques include: • Assessments • Checklists
Review the WBS, plan, and schedule for opportunities.	
Identify the benefits and costs of each opportunity.	
Assign priorities to each opportunity based on the benefit and cost.	
Rank opportunity according to priorities.	
Review assigned opportunity priorities with affected stakeholders and get agreement.	
Develop the risk and opportunity analysis report.	Include a list of prioritized risks or opportunities.

Example Work Products

Example Work Products	Further Explanation
Identified risks or opportunities	These include: • Risk impact and probability of occurrence • Opportunity benefit and cost
Risk or opportunity priorities	
Risk and opportunity analysis reports	

RSK 2.2

Required Practice Information

Practice Statement

Monitor identified risks or opportunities and communicate status to affected stakeholders.

Value

Enables timely corrective or leveraging actions to maximize the likelihood of achieving objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Review risks or opportunities periodically, including changing conditions. Reexamination may uncover risks or opportunities that were previously overlooked or that did not exist when the identification and priorities were last updated.

Example Activities

Example Activities	Further Explanation
Periodically review risks or opportunities.	Review risks or opportunities in the context of status, circumstance, and past or planned activities.
Update risks or opportunities as additional information becomes available.	Risks or opportunities may be revised when: • New risks or opportunities are identified • Corrective or leveraging actions are taken • Risks are retired • Opportunities are accepted • Circumstances change significantly
Communicate the risk or opportunity status to affected stakeholders.	Examples of risk or opportunity status include a change in the: • Probability of occurrence • Costs or benefits • Priority • Impact on objectives

Example Work Products

Example Work Products	Further Explanation
Records of risk or opportunity monitoring	
Updated risks or opportunities	

Level 3

RSK 3.1

Required Practice Information

Practice Statement

Identify and use risk or opportunity categories.

Value

Organizes risks or opportunities to focus attention on uncertainties that will impact the achievement of objectives.

Additional Required Information

Include the name and a brief description for each category.

Explanatory Practice Information

Additional Explanatory Information

The use of categories helps provide structure and efficiency for risk or opportunity identification and analysis.

Identify categories over time to uncover changing circumstances that affect the ability to meet objectives. As the work progresses, identify additional categories of risks or opportunities.

Categorization organizes related risks or opportunities to support the efficient and effective use of resources.

Example Activities

Example Activities	Further Explanation
Identify risk or opportunity categories.	Consider internal and external categories, such as: • Work activities • Types of processes used • Types of resources used • Types of solutions produced • Regulations and laws • Work management uncertainties, e.g., related to contracts, budget, schedule, quality, resources • Technical performance uncertainties, e.g., related to quality characteristics, reliability
Organize risks or opportunities according to defined categories.	Related or equivalent risks or opportunities can be organized for efficient handling.

Example Work Products

Example Work Products	Further Explanation
Categories list	
Categorized risks or opportunities	

RSK 3.2

Required Practice Information

Practice Statement

Define and use parameters for risk or opportunity analysis and handling.

Value

Maximizes the likelihood of cost-effectively achieving objectives.

Additional Required Information

Parameters include:

- Probability of occurrence
- Impact
- Expected outcomes

Explanatory Practice Information

Additional Explanatory Information

Use defined parameters to:

- Determine the severity of the risk
- Estimate the benefit of the opportunity
- Prioritize the actions required for planning
- Select the opportunities to pursue

Parameters for evaluating, categorizing, and prioritizing include:

- Probability, e.g., likelihood of occurrence
- Impact, e.g., consequence and severity of occurrence
- Thresholds to trigger actions
- Expected benefit from opportunity
- Expected cost of opportunity

Parameters are often used in combination when prioritizing risks or opportunities. For example:

- Set priorities by multiplying probability times impact
- Calculate an opportunity value by subtracting the expected cost from the expected return.

Example Activities

Example Activities	Further Explanation
Identify a relative priority for each risk or opportunity based on assigned parameters.	
Define thresholds to trigger actions for selected risks or opportunities.	Identify which thresholds require a: • Mitigation activity • Contingency plan • Opportunity action plan
Prepare and perform assessments for selected risks.	Risk assessments focus the components of the risk strategy on analyzing a specific situation and risk. Examples of risk assessments include: • Exploitation • Malicious uses • Safety • Regulations and laws • Cyber security
Prepare and perform opportunity assessments.	Examples of opportunity assessments include: • Cost benefit analyses • Future needs analyses

Example Work Products

Example Work Products	Further Explanation
Risk or opportunity evaluation, categorization, and prioritization parameters	
List of risks or opportunities and their assigned priority	
Risk or opportunity assessment results	

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Identify potential hazards associated with any subsystem interfaces and faults. Assess the risk associated with an integrated system design, including software and subsystem interfaces.

Leverage safety functional groups to assess the severity and probability of safety risks associated with each identified hazard to determine the potential negative impact of the hazard. For example, consider potential impacts to:

- Personnel
- Facilities
- Equipment
- Operations
- Public
- Environment
- System or solution

If safety risks increase beyond tolerance limits, then identify short-term and long-term actions to reduce the risk to an acceptable level. Depending on the product and hazard scenario, this may take the form of operational limitations, usage restrictions, in-service tests/inspections, or design/manufacturing changes.

RSK 3.3

Required Practice Information

Practice Statement

Develop and keep updated a risk or opportunity management strategy.

Value

Avoids problems and leverages opportunities to increase the likelihood of achieving objectives.

Additional Required Information

A risk or opportunity management strategy includes:

- A description of how risk or opportunity management will be performed
- Scope
- Methods and tools
- Risk or opportunity categories
- Risk or opportunity parameters
- Thresholds
- Risk mitigation techniques
- Opportunity leveraging techniques
- Risk measures
- Opportunity measures, including:
    - Costs
    - Benefits
- Frequency of risk monitoring or reassessment

Explanatory Practice Information

Additional Explanatory Information

Develop the strategy early and record it in the organization's or project's risk or opportunity management plan. Keep the strategy updated throughout the project. The strategy is used to guide risk and opportunity management activities. The strategy may be included as part of the project plan.

The risk or opportunity management strategy may also include:

- The domain of interest
- Boundaries, inclusions, and exclusions
- Interactions, dependencies, and relationships between the solution or work and external factors
- Methods to be used to derive:
  - Risk or opportunity categories
  - Impact and likelihood of occurrence
  - Benefit and cost of the opportunity
  - Acceptance criteria
- Conditions that initiate a review and potential update to the risk and opportunity management strategy

Example Activities

Example Activities	Further Explanation
Develop, record, and keep updated a risk or opportunity management strategy.	
Review the risk or opportunity management strategy with affected stakeholders.	Use the review to gather input to: • Improve the strategy • Generate acceptance

Example Work Products

Example Work Products	Further Explanation
Risk or opportunity management strategy	

RSK 3.4

Required Practice Information

Practice Statement

Develop and keep updated risk or opportunity management plans.

Value

Minimizes the impact of risks and maximizes the benefits of opportunities for achieving objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Generate mitigation or contingency plans for selected risks. Mitigation plans describe how to reduce the likelihood or impact of risks. Contingency plans deal with the impact of problems that can occur despite mitigation attempts.

Options for mitigating risks include:

- Avoiding
- Controlling
- Transferring
- Monitoring
- Accepting

Options for leveraging opportunities include:

- Creating
- Exploiting
- Transferring
- Monitoring
- Enhancing
- Accepting

More than one approach to managing a risk or opportunity can be used.

Mitigation and contingency plans can include:

- Rationale
- Cost benefit analysis
- Risk acceptance criteria
- Schedule or Period of Performance (PoP) for each risk and opportunity management activity
- Resource reserves to respond to disruptive events
- Lists of available backup equipment
- Backups to key personnel
- Plans for testing emergency response systems
- Posted procedures for emergencies
- Distributed lists of key contacts and information resources for emergencies
- Actions to be taken

Generate leverage plans for selected high-priority opportunities. These plans describe how to maximize the benefit of an opportunity.

Leveraging involves performing actions that maximize the benefits of an opportunity without increasing the cost beyond the benefit. Typically, leveraging adds a relatively small amount of cost while yielding a relatively high level of benefit.

Leveraging plans include:

- Cost benefit analyses
- Potential for success analyses
- Preparation activities
- Actions required to leverage opportunity

This activity can result in the discovery of new opportunities that can require replanning and reassessment.

Example Activities

Example Activities	Further Explanation
Develop mitigation plans for selected risks and contingency plans in the event their impacts are realized.	
Develop leverage plans for selected opportunities to increase the likelihood that their impacts are realized.	
Review plans with affected stakeholders.	Use the review to gather input to: • Improve the plan • Generate acceptance

Example Work Products

Example Work Products	Further Explanation
Risk management plans	Includes: • Mitigation • Contingencies
Opportunity leveraging plans	
Updated plans and status	

Related Practice Areas

Planning (PLAN)

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Evaluate identified safety risks and consider appropriate mitigations:

- To address the minimum safety tolerance limit as defined by requirements and the system user or customer
- To ensure compliance with laws, e.g., country, federal, and state where applicable, regulations, executive orders, treaties, and agreements

Following initial hazard identification, identify risk mitigation strategies and evaluate the expected effectiveness of each alternative or method. Risk mitigation is often an iterative process for eliminating or reducing risk to the lowest acceptable level within the constraints of operational effectiveness and suitability, time, and cost.

Collect information and data needed to conduct a mitigation analysis. For example, perform a Preliminary Hazard Analysis (PHA) to conduct an initial safety risk assessment of a concept or system. The analysis should include consideration of the following concepts:

- Hazardous components, e.g., fuels, propellants, lasers, radio transmitters, explosives, toxic substances, hazardous construction materials, pressure systems, and other energy sources
- Interface considerations among various elements of the solution or system, e.g., material compatibilities, electromagnetic interference, inadvertent activation, fire/explosive initiation and propagation, and hardware and software controls; and to other systems when in a network or system-of-systems architecture
- Operating environment constraints, e.g., drop; shock; vibration; extreme temperatures; noise; exposure to toxic substances; health hazards; fire; electrostatic discharge; lightning; electromagnetic environmental effects; and radiation, including ionizing, non-ionizing, laser, and radio-frequency
- Process constraints for operations, tests, maintenance activities, diagnostics, and emergency procedures. Other considerations include:
    - Environmental impacts
    - Human factors engineering
    - Human error analysis of operator functions, tasks, and requirements
    - Effect of factors such as equipment layout, lighting requirements, potential exposures to toxic materials, and effects of noise or radiation on human performance
    - Explosive ordnance render-safe and emergency disposal procedures
    - Life support requirements and safety implications in manned systems, including crash safety, egress, rescue, survival, and salvage

- Developed infrastructure, property, installed equipment, and support equipment, e.g., provisions for storage, assembly, and checkout; proof testing of hazardous systems/assemblies that may involve toxic, flammable, explosive, corrosive, or cryogenic materials or wastes; pollution, radiation, or noise emitters; and electrical power sources
- Natural infrastructure, e.g., land use, water resources, air quality, geology and soils, biological resources, cultural resources, hazardous materials, solid and hazardous waste, environmental noise, and aesthetic and visual resources
- Environment, Safety, and Occupational Health (ESOH) controls, safeguards, and possible alternate approaches, e.g., interlocks; system redundancy; fail safe design considerations using hardware or software controls; subsystem protection; fire detection and suppression systems; personal protective equipment; heating, ventilation, and air-conditioning; noise or radiation barriers; and pollution control equipment
- Malfunctions of the system, subsystems, or software; specify each malfunction, determine the cause-and-result sequence of events, assess the hazard, and develop appropriate specification or design changes

When identifying mitigations, consider and assess the likelihood of potential hazards leading to accidents, and the associated impact.

The likelihood may be assessed qualitatively, e.g., frequent, rare, or extremely rare; or quantitatively, e.g., 1 occurrence in 10 years, 1 occurrence every 10,000 operations, or 1 occurrence every 100 missions. A quantitative measure of the likelihood of a potential accident is based on the analysis of the hazards for which that accident is a possible consequence.

When systematic failures are seen as contributing to the likelihood of an accident, e.g., where software failures may cause the accident, quantitative analysis is generally seen as inappropriate. Instead, the process is often reversed to set safety targets for system failure rates.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Consider all risks in planning activities. Identify risks or opportunities from multiple perspectives, e.g., acquisition, technical, management, operational, supplier agreement, industry, support, end user. Consider applicable regulatory and statutory requirements, e.g., safety and security, while identifying risks. As the work evolves, revise risks based on changed conditions.

There are many risks associated with acquiring solutions through suppliers, e.g., the stability of the supplier, the ability to maintain sufficient insight into the progress of their work, the supplier’s capability to meet solution requirements, and the skills and availability of supplier resources to meet commitments.

Analyze the process and solution level measures and associated thresholds to identify where the organization is at risk of not meeting thresholds. These measures are key indicators of risk.

RSK 3.5

Required Practice Information

Practice Statement

Manage risks or opportunities by implementing planned risk or opportunity management activities.

Value

Reduces unforeseen occurrences that impair ability to achieve objectives and increases business value by leveraging opportunities.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

This activity is more than simply monitoring risks and opportunities. It uses analyses, plans, triggers, and thresholds to anticipate actions needed to minimize risk impact or leverage opportunities to improve project functioning.

Example Activities

Example Activities	Further Explanation
Manage risks or opportunities using the risk or opportunity management plans.	Continue to manage risks or opportunities after mitigation, contingency, or leveraging activities have started. Measurement can provide valuable insight into the risk or opportunity management activities.

Example Work Products

Example Work Products	Further Explanation
Updated status	Includes status of mitigation, contingency, and leveraging plans.